Vineet Madasseri Payyappalli

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EDUCATION

University at Buffalo, The State University of New York

Doctor of Philosophy in Industrial Engineering (Operations Research), *May 2019 (expected)* GPA: 3.87/4.00

Master of Science in Industrial Engineering (Operations Research), *May 2015* GPA: 3.86/4.00

National Institute of Technology (NIT) Calicut, India

Bachelor of Technology in Mechanical Engineering, *May 2009* GPA: 7.09/10.00

RELEVANT SKILLS

Data analysis: R, SQL, RShiny, Spark

Machine learning: keras, TensorFlow, scikit-learn

Statistics: Experimental design, A/B testing

Programming: Python, Java

Python/R packages: pandas, numpy, dplyr, ggplot2

Optimization: CPLEX, docplex, Matlab

INDUSTRY EXPERIENCE

Data Science Intern, WestRock, *Atlanta, GA* *May 2018 – August 2018*

Performance improvement of Supply Chain Optimization (SCO) model **[CPLEX, Python, R]**

- Reduced SCO model's solve time by 15%, using data analysis to identify bottlenecks in solution process.
- Developed Python scripts for automating tests and for implementing distributed parallel computing.
- Designed and conducted a statistical experiment and performed A/B testing of two software versions.
- Conducted machine network's data analysis and clustering using R and Python, to help parallelization.

Team Leader & Consultant, Meals on Wheels for WNY, *Buffalo, NY* *Mar 2017 – Oct 2018*

Warehouse improvement project **[MS Excel]**

- Led a team of 6 students to develop a VBA tool that quantified warehouse capacity and utilization.
- Developed a check-list that reduced incorrect meal deliveries by 73.3%, during pilot test.

Manufacturing & Quality Control Engineer, Areva/Alstom T&D. *Chennai, India* *Jul 2009 – Jun 2013*

- Reduced assembled products inventory by 75% (11.438 million INR) in 2 years. **[MS Excel]**
- Proposed statistical safety stock calculations to improve inventory turns ratio about three-fold (from 3 to 10) and to reduce risk of stock-outs for critical components. **[MS Excel]**

RESEARCH EXPERIENCE

Research Assistant, University at Buffalo, *Buffalo, New York* *Aug 2014 – Present*

Spatio-temporal fire risk prediction and analysis **[Python]**

- Working on predicting fire risk at spatial and temporal levels.
- Wrangled and merged complex datasets: NFIRS (400 million records); U. S. Census, weather data.

Integer programming model for soccer clubs to optimize player transfers **[CPLEX, Python, R]**

- Developed Python web-scraper to collect data on soccer players and clubs (200,000 rows).
- Created and solved an integer program to optimize investments and maximize the club's market value.

Resource allocation model to mitigate losses from structural fires **[R, Matlab, RShiny, Python]**

- Conducted non-linear regression to estimate investment effectiveness and vulnerability parameters.
- Developed and solved constrained non-linear models to minimize losses from fires.

Game-theoretic cyber information sharing model using particle swarm optimization **[Matlab]**

- Created and solved two-stage constrained non-linear model.
- Implemented gradient-based algorithms and multi-objective particle swarm heuristics.

Predicting the occurrence of illegal fishing in the Gulf of Mexico **[R]**

- Implemented linear regression, logistic regression, kNN, LDA, QDA, rpart, and random forest in R.
- Predicted occurrence of illegal fishing in the Gulf of Mexico, using U.S. Coast Guard's data.

PROFESSIONAL ACTIVITIES

President of Student Chapter of INFORMS, University at Buffalo *Aug 2017 – May 2018*

- Organized a record 36 events, leading to INFORMS awarding the chapter Summa Cum Laude.
- Identified, developed, and completed student-run projects with not-for-profit organizations.
- Served as President-Elect (Sep 2016 - Aug 2017), and as Vice President (Aug 2014 - Sep 2016).

AWARDS & ACHIEVEMENTS

Finalist, The Principal Cup: An OR/MS Anlaytics Challenge [Python, R] *Sep 2018*

- Developed classification models (random forest multi-output classifier in Python; naive Bayes in R) that help portfolio managers make decisions on whether to sell, buy, or hold a stock after a shock event.
- Used Principal Financial's events data (106,740 events with 38 features).

Finalist, Dow Big Data Challenge [R] *Apr 2016*

- Wrangled Dow's 18-month transactional orders data (structured data; more than 2 million rows).
- Implemented ARIMA, hidden Markov, and exponential smoothing models for time-series prediction, in R.
- Predicted likelihood of specific customers ordering a given product in a given month, including the quantity.

Excellence in Research, Scholarship and Creativity Certificate of Excellence *April 2018*

- Awarded by the Office of the Graduate School, University at Buffalo, in recognition for exemplary work in graduate research and scholarship.

Runner-up for Decision Analysis Society Best Student Paper Award *Oct 2017*

- Selected as a runner up (from among 20 applicants) for the best student paper in 2016.

PUBLICATIONS

Payyappalli, V. M., & Zhuang, J. (2019). A data-driven integer programming model for soccer clubs' decision making on player transfers. Environment Systems and Decisions, *forthcoming*.

Behrendt, A., Payyappalli, V. M., & Zhuang, J. (2019). Modeling the Cost Effectiveness of Fire Protection Resource Allocation in the United States: Models and a 1980-2014 Case Study. Risk Analysis, *forthcoming*.

Guan, P., Zhang, J., Payyappalli, V. M., & Zhuang, J. (2018). Modeling and Validating Public–Private Partnerships in Disaster Management. Decision Analysis, 15(2), 55-71.

Payyappalli, V. M., Zhuang, J., & Jose, V. R. R. (2017). Deterrence and Risk Preferences in Sequential Attacker–Defender Games with Continuous Efforts. Risk Analysis, 37(11), 2229-2245.

Zhuang, J., Payyappalli, V. M., Behrendt, A., & Lukasiewicz, K. (2017). Total Cost of Fire in the United States. Fire Protection Research Foundation.